Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
Prof. Dr. Sofia C. Olhede

Practice Exam — Week 4: Parametric Estimation — Yule–Walker & Least Squares

Duration: 60 minutes · No calculator · Answer on paper

Name: ______________________ Surname: ______________________

SCIPER (Student Card Number): ______________________

Exercise	**Points**
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2.

Problem 1 (5 points)

Let $\{X_t\}$ be a mean-zero causal AR(2) process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$.

(a) Starting from the model equation, derive the *Yule–Walker equations*: multiply by X_{t-k}, take expectations, and explain precisely where *causality* is used to kill the noise cross-term. State the resulting linear system $\boldsymbol{\gamma} = \Gamma \boldsymbol{\phi}$ and the variance identity for σ^2. (2 pts)

(b) A series gives the sample autocovariances $\hat{\gamma}_0 = 5$, $\hat{\gamma}_1 = 3$, $\hat{\gamma}_2 = 1$. Compute the Yule–Walker estimates $\hat{\phi}_1, \hat{\phi}_2$ and $\hat{\sigma}^2$. (2 pts)

(c) Verify that the fitted model of part (b) is causal (stationary) by locating the roots of its AR polynomial. (1 pt)

Problem 2 (5 points)

A series of length $n = 150$ is modelled as a mean-zero AR(3) process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \phi_3 X_{t-3} + \varepsilon_t$. The estimated autocorrelations are

$$\hat{\rho}_1 = 0.8, \qquad \hat{\rho}_2 = 0.6, \qquad \hat{\rho}_3 = 0.4,$$

and the sample variance is $\hat{\gamma}_0 = 10$.

(a) Explain why the Yule–Walker fit can be written entirely with the *correlation* matrix, $\hat{\boldsymbol{\phi}} = \bar{\Gamma}_3^{-1}\hat{\boldsymbol{\rho}}_3$, i.e. why $\hat{\gamma}_0$ cancels in the system for $\boldsymbol{\phi}$. Write out the 3×3 system explicitly. (1.5 pts)

(b) Solve the system for $\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3$ (e.g. by Gaussian elimination) and verify your solution by substituting it back into the three equations. (2.5 pts)

(c) Compute the innovation-variance estimate $\hat{\sigma}^2$ using $\hat{\gamma}_0 = 10$. (1 pt)

Problem 3 (5 points)

This problem treats the two faces of least squares.

(a) *(MA via residual reconstruction.)* A mean-zero MA(1) process $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$ is observed at $X_1 = 0$, $X_2 = 2$, $X_3 = -1$. Reconstruct the innovations $\hat{\varepsilon}_1, \hat{\varepsilon}_2, \hat{\varepsilon}_3$ as functions of θ (state your initialisation), write the least-squares criterion $S(\theta)$, and find $\hat{\theta}$. Is the fit invertible? (2.5 pts)

(b) *(Forward least squares for* AR(2).*)* For the five observations $x_1, \dots, x_5 = 1, 2, 0, -1, -2$ (already mean-zero), write the forward design $\mathbf{X}_F = F\boldsymbol{\phi} + \boldsymbol{\varepsilon}$ for an AR(2) fit, form the normal equations $F^\top F\,\hat{\boldsymbol{\phi}}_F = F^\top \mathbf{X}_F$, and solve for $\hat{\boldsymbol{\phi}}_F$. Then report $\hat{\sigma}_F^2$ using the correct degrees of freedom. (2 pts)

(c) State, in one sentence each, (i) why the *backward* least-squares estimator $\hat{\boldsymbol{\phi}}_B$ estimates the *same* coefficients $\boldsymbol{\phi}$, and (ii) whether a least-squares AR fit is guaranteed to be stationary. (0.5 pt)

Problem 4 (5 points)

(Revision of Week 3.) Consider the ARMA$(2,1)$ process

$$\left(1 - \tfrac{3}{4}B + \tfrac{1}{8}B^2\right)X_t = \left(1 + \tfrac{1}{3}B\right)\varepsilon_t.$$

(a) Factor the AR polynomial $\Phi(z) = 1 - \frac{3}{4}z + \frac{1}{8}z^2$, locate its roots, and confirm the process is causal. Then check invertibility from $\Theta(z) = 1 + \frac{1}{3}z$. (2 pts)

(b) Using a *partial-fraction* expansion of $G(z) = \Theta(z)/\Phi(z)$, find a closed form for the MA(∞) weights ψ_j in $X_t = \sum_{j\geq 0} \psi_j \varepsilon_{t-j}$, and report ψ_0, ψ_1, ψ_2. (3 pts)